Decision Tree Classification Flow

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1 Step 1 System Entropy

The first step for any CART is to find the entropy of the target columns, this value is called system entropy.

1.1 Calculations and logic assesment.

First Single out the target columns, find out the number of classes and value counts for each class.

$$\begin{array}{ccc} a & \longrightarrow & 200 \\ b & \longrightarrow & 211 \\ c & \longrightarrow & 192 \\ \vdots & \longrightarrow & \vdots \\ n & \longrightarrow & x \end{array}$$

Find the probabilities for every class, say for class a:

$$P(a) = \frac{200}{\text{Total Rows Number}}$$

Compute the system entropy using the formula:

$$H(s) = -\sum_{i=1}^{n} P(i) \log_2(P(i))$$

2 Step2: Finding the Splitting criteria

A loop is ran where every feature column is targetted and its Information Gain(I.G) is calculated, the column with higher Information gain in order are selected as priority splitting criteria.

The approach for findng I.G differs based on the type of the data the targetted feature column holds. The approach differs by evaluating the type of data on hand, if the column consists of values that can be exacted/repeated multiple times(similar to discrete), or no consistent repition(continous in nature).

2.1 Discrete nature of Data

These columns of data can be one hot encoded. Similar to the target column these feature columns are also supposed to have unique elements and counts associated to them. Then we perform **Binary Subset Split**. In this split as the name suggests, only two category is conisdered. Say one of the Category is A then the other category becomes A or $U - A$. Then we calculate the **Weighted Entropy**. This is repeated for every Category presented in the feature column.

2.1.1 I.G for a Discrete nature column.

Let's assume a feature column with three distinct features in it (0,1,2) after encoding from strings and target column with target vairables as (A,B,C). First we compute, the metrics such as counts of mapping such as counts of a feature varaible and target variable.

$$\begin{array}{ccc} 0_A & \longrightarrow & 344 \\ 1_A & \longrightarrow & 221 \\ 2_A & \longrightarrow & 343 \\ 0_B & \longrightarrow & 222 \\ \vdots & \longrightarrow & \vdots \\ 2_C & \longrightarrow & 212 \end{array}$$

Now, lets compute the binary split of 0 and 0^{C}.

we have 0 and $0^{\complement}$, our goal is to find the Weighted entropy of a binary split, so the formual becomes.

$$H_w(X) = \sum_{i=1}^{n} P(0_i)H(0_i) + P(0_i^{\complement})H(0_i^{\complement})$$

The probabilities are used as weights and the notations are tried to be summarized below:

$$\begin{aligned}
n &= \text{Number of Unique classes in target column} \\
P(0_i) &= \text{weight 1 (Probability of 0 in feat col)} \\
P(0_i^{\complement}) &= \text{ weight 2 (Probability of } 0^{\complement} \text{ in feat col)} \\
H(0_i) &= \text{Entropy of 0th component of binary split} \\
H(0_i^{\complement}) &= \text{Entropy of 1th component of binary splilt}
\end{aligned}$$

Entropy of the binary splits:

$$\begin{aligned}
H(0_i) &= P(0_A)\log 2(P(0_A)) + P(0_B)\log 2(P(0_B)) + P(0_C)\log 2(P(0_C)) \\
H(0_i^{\complement}) &= P(0_A^{\complement})\log 2(P(0_A^{\complement})) + P(0_B^{\complement})\log 2(P(0_B^{\complement})) + P(0_C^{\complement})\log 2(P(0_C^{\complement}))
\end{aligned}$$

We repeate this for the set (1 ,$1^{\complement}$) and (2, $2^{\complement}$), and find the I.G by subtracting the weighted entropy for each set.

2.2 Continuos nature of Data

To find the splitting criteria in a column of continous nature modern CART models use a combined approach of **Greedy Approach and Histogram Binning**. Let's look at them individually.

2.2.1 Greedy Approach

In Greedy Approach, we leave out no possible candidate for splitting, this is definitely a compute heavy approach. Lets imagine a continous feature columns and it's corresponding target column with three specific targets A,B and C.

Feat Col	Targ Col
21	B
14	C
42	C
19	A
36	A
21	B
14	A
59	B
68	C
75	C

Table 1: Feat and Targ cols

This apporach is supposed to work as, find the mean between every sorted unique elements in the feature columns and find the Information Gain for that mean and find the one with the Highest Information Gain and use it as the splitting criteria.

Sorted Feats	Targ Col	Upper and Lower end
14	A	–
14	C	14-19
19	A	19-21
21	B	–
21	B	21-36
36	A	36-42
42	C	42-59
59	B	59-68
68	C	68-75
75	C	Upper right end

Table 2: Feat and Targ cols

We obviously ignore the repeated values as they serve as the same bounding parameter while calculating the mean, but the associated value in the target column should be recorded for entropy calculation.

- **Lopping**
  We run the loop $k-1$, where k is the number of rows in the given data.

- **Entropy Calculation**
  For every mean we look for the

  $H_L = \sum P(A_L) \log 2(P(A_L)) + P(B_L) \log 2(P(B_L)) + P(C_L) \log 2(P(C_L))$

  and

  $H_R = \sum P(A_R) \log 2(P(A_R)) + P(B_R) \log 2(P(B_R)) + P(C_R) \log 2(P(C_R))$

  where,

$$\begin{aligned}
P(A_L) &= \text{Porbability of A in left side of mean} \\
P(B_L) &= \text{Porbability of B in left side of mean} \\
P(C_L) &= \text{Porbability of C in left side of mean} \\
P(A_R) &= \text{Porbability of A in right side of mean} \\
P(B_R) &= \text{Porbability of B in right side of mean} \\
P(C_R) &= \text{Porbability of C in right side of mean}
\end{aligned}$$

- Weighted Entropy

$$H_w(X) = \sum P(L)H(L) + P(R)H(R)$$

  where,

$$\begin{aligned}
P(L) &= \text{Porbability of a entity being on left of mean} \\
P(R) &= \text{Porbability of a entity being on right of mean}
\end{aligned}$$

3 Programming Logic

Decision is one of the classic problem which uses **Recursion** to find the splitting criteria and build untill a base criteria is met.

Programming using Object Oriented approach helps simplifying any kind of programming logics, as it defines the ownership and state properly so it is easire to bebug and make changes.

But before writing OOP logic, it must be clear about the ownership of the state and components in use.

3.1 Components and Ownership

Lets classify the Components hierarchialy, namely from **Classes**, as it sits on the top of the objects and serves as a blue print.

3.1.1 Classes

Name	Nature	Utility	Attributes	Methods
DecisionTree (DT)	Independent	Provides base class for Classification and Regression Tasks.	depth, max_depth, min_entropy	build_tree
Node	Independent	Provide branching/ endpoint in the Tree.	value, left, right, feature, threshold	None
CriteriaFinder (CF)	Independent	Compute mathematical aspects for the DT	sys_entropy, feat_entropy, ig, probs	find_criteria, cal_probs, cal_entropy, cal_ig
DecisionTreeClassifier (DTC)	Inherited DT	DT Specialized for handling classification tasks.	None	None
DecisionTreeClassifier (DTC)	Inherited DT	Supposed to hold the data that arrives for being	num cols	shuffle

Table 3: Independent Components

4 Histogram Binning, Move to a different document

This is the more of compute friendly approach where, we divide the data in bins and try singling out one of the bin, which is able to point out the maximum information for that column, the bins is either averaged or one of the end is taken as the criteria. This apporach is also called **Histogram Binning**.